

Alex — Tier 1 sequential model outline

Role of this document

This note is the **advisor-ordered spine**: each section follows the sequence Alex emphasized (minimal model → assumptions → data → fitting → what we extract). It **does not replace** existing work; it points to it.

Paper Directions 11 (pool pre-sorting & promotion rules) is §6A — separate from the 538 empirical ladder (§4–§6). Read §6A before treating 538 as addressing “how teams are built.” **Printable design note:** [sports/documents/Tier1_Presorting_Design_Note.md](#).

Stable references (do not fork their content here):

- [Tier1_Briefing_Outline.md](#) — full technical detail, equations, and data column map.
- [Tier1_Narrative_Outline.md](#) — narrative arc and voice.
- [sports/documents/tier_1_roadmap.md](#) — execution contract for 535 / panel columns.
- [sports/538_alex_tier1_model_and_fit.ipynb](#) — notebook that should **mirror this section order** for code + outputs (537 stays the simulation lab). **Cell-level map:** §10 below (update when 538 changes).
- [sports/documents/tier_1_roadmap.md](#) — 535 execution contract (pipeline steps, df columns, CELL 0 switches for 535_sports_tier_1.ipynb). Not superseded by 538; 538 reuses the same pipeline modules with a slimmer cell ladder.
- **Wang-style cross-domain template** (same *logical* steps as this outline): [Vector_Questions_and_Modeling_Thoughts.md](#) §11; mechanics of the Yin–Wang failure/success paper: [wang_paper_model.md](#).

1. Unit of modeling

- State the primary unit (i, j, t) and the empirical row (player–season / player–team–season).
- One sentence: local pool vs global advancement.

Fill from [Tier1_Briefing_Outline.md](#) §1; keep only what you will say aloud.

2. Minimal model objects (headline only)

- Outcome Y (global advancement).
- Amalgamated local environment L_{ijt} (Tier 1 object); first proxy Q .
- Global selection capacity Λ_t .
- Role of own performance / ability proxy A_{ijt} (baseline, not the mechanism headline).

Detail table: [Tier1_Briefing_Outline.md](#) §2–§3.

3. Minimal assumptions

- Local vs global scarcity framing.
- Leave-self-out for peer-based Q .
- Variable domains (binary Y , standardized L for fitting, etc.).
- What Tier 1 **defers** (sorting, dynamics, full causal ID). **How pools are formed** is the PD11 generative agenda — see §6A; §4–§6 **take pools as given** in the data.

Bullets only here; expand in manuscript later.

4. Map model to data

One table: object → meaning → `df` column / construction step (`530` / panel load + Tier 1 cols).

Canonical table: `Tier1_Briefing_Outline.md` §4.

In `538`: CELL 1 (`CFG`, panel path) → CELL 2 (`load_panel`, `apply_perf_metric_for_analysis`, legacy `poolq_loo`) → CELL 3 (`add_tier1_mechanism_variables`, `ALEX_L_COL` from `PRIMARY_POOL_MODE`).

5. Primary estimand

- **Main object to report:** turning point L^* (or Q^* in raw units), not only linear coefficients.
- One sentence on why (competing margins / inverted-U readout).

Formulas: `Tier1_Briefing_Outline.md` §7.

6. Fitting plan (ordered)

1. Descriptive bins (no model). → `538` CELL 4 (table), CELL 4A (binned line; optional if CELL 5 overlay is enough).
2. Transparent quadratic spec in L (LPM) + controls as needed. → `538` CELL 5 (OLS on z-scored L , L^2 , A ; binned means + fitted curve).
3. Binary response (logit/probit) for the same index structure. → `538` CELL 6 (logit; same index; L^* from linear part).
4. Robustness: FE, clustering, alternative L proxy, one decomposition term at a time. → `538` CELL 7+ (placeholder).

Exploratory (not Alex §6 step 1; from Paper Directions 11 meeting): mean peer Q × LOO peer perf SD vs draft. → `538` CELL 4B (hexbin + Q histogram), CELL 4C (3D surface; optional). Always uses `congestion_quality` on x even when `PRIMARY_POOL_MODE='crowding'`.

Expand: `Tier1_Briefing_Outline.md` §5–§6.

6A. Paper Directions 11 — pool formation, promotion rules, and where they live

Source: `transcripts/20260515_Paper_Directions_11_otter_ai_transcript.pdf` (2026-05-15).

Do not confuse with §6: §6 is the Wang-style empirical ladder on realized pools. PD11 is about questioning the rules that put people in pools and that turn pool context into promotion — especially **pre-sorting / team building** in a **generative** layer (planned `538`; `537` frozen), not re-running more parameter grids.

What PD11 is *not* asking for (here)

- Replacing §6 with new bins/LPM/logit specs alone.
- Calibrating the generative model by **copying observed college team means** — Alex: misleading for a **multi-domain** minimal story (forensic checks on overlap/inequality across domains are fine).
- Treating `538` as if it had already solved pool formation — it has not (see **Where each thread lives** below).

Three threads (meeting order of emphasis)

#	Thread	Question	Priority in code
A	Pre-sorting / how teams (pools) are built	How do we assign individuals to local pools so pools overlap in talent, are not rigidly ordered slices, and # pools » # quantile bins ?	<code>538</code> (planned generative cells + <code>tier1_sim_config.py</code>). <code>537</code> frozen (legacy sort-and-chop in <code>assign_pool_ids</code>). Forensics: <code>530</code> CELL 8–9.
B	Pool mean × pool dispersion	Does advancement depend on LOO mean peer ability <i>and</i> LOO SD of peer ability (crowding vs distinction)?	<code>538</code> CELL 4B, 4C (exploratory EDA). Columns: <code>congestion_quality</code> , <code>peer_perf_sd_loo</code> (<code>tier1_mechanism_vars.py</code>).
C	Promotion score vs local rank	Should promotion weight gap to pool mean ($A_i - \bar{A}_{\text{pool}}$) rather than rank within pool (same rank whether peers are tight or spread)?	<code>537</code> first — e.g. <code>ADDITIVE_LOCAL_RANK_WEIGHT</code> / score mode in <code>sim_config.py</code> , Cell 10 playground. Then revisit A if shape does not move.

Alex’s implementation order (transcript): tweak **C** in existing simulation machinery first (easy); if insufficient, rebuild **A**; use **B** in data to see whether dispersion belongs in the story.

Thread A — pre-sorting (generative pool assignment)

Problem with the old rule (still in `537` for assortative pools):

1. Draw abilities for all i .
2. Sort (optionally add Gaussian ϵ to a **sorting signal**).
3. **Chop** the sorted list into J equal-sized pools → **hard, ordered** teams with little realistic overlap.

Critique (PD11): teams look like **non-overlapping talent intervals**; ϵ -smearing is **too linear** for **heavy tails** (elite player on weak team occasionally); **# teams must be much larger than # quantile bins** used in EDA.

Directions discussed (generative — pick one family to prototype in 538):

1. Target means, then soft assignment (preferred framing)

- Draw a **target mean ability** T_j for each pool j (e.g. J draws from $\text{Uniform}[0, 1]$ or another spread).
- Assign i with ability α_i to pool j with probability **↑ as α_i is close to T_j** (Gaussian kernel around T_j , or heavier tail).
- Yields **overlapping “talent windows”** across pools — some tight, some spread out.

2. Sequential / preferential assignment

- Start with even assignment probabilities; **update** as pools fill (Barabási-style “rich get richer”) — another way to avoid rigid slices.

3. Heavy-tailed reassignment noise

- Charles: ϵ from a **heavy-tailed** distribution so elites occasionally land in weak pools but usually sort high — closer to rare “superstar on weak program” events.

Assortativity knob: width of the assignment kernel (or attachment strength) controls how **assortative** mixing is — still need many pools vs few bins.

Empirical basketball: pools are **observed** (team_id, season) after real sorting (recruiting, transfers, minutes). §4–§6 **measure** LOO Q and peer SD **within** that realization; they **do not** re-simulate assignment.

Thread B — mean × SD (empirical diagnostic)

- **x:** LOO mean peer performance → congestion_quality (same object as primary Q / ALEX_L_COL when PRIMARY_POOL_MODE='quality').
- **y:** LOO sample SD of teammate perf → peer_perf_sd_loo.
- **Color / z:** mean Y_draft (or promotion rate in other domains).

Hypothesis (talking point): low peer SD → **crowding** → stronger inverted-U; high peer SD → easier **distinction** → weaker crowding read. **Ranking alone drops SD information** — motivates B alongside mean-only Q .

538 note: 4B/4C fix x to congestion_quality even when PRIMARY_POOL_MODE='crowding' (crowding toggle affects §6 ladder cells, not this PD11 plot).

Thread C — promotion rule (simulation)

Old composite (conceptual): $w \cdot \text{local_rank} + (1 - w) \cdot A_i$.

PD11 push: weight should reflect **how much better than the pool mean**, not “#1 in pool” regardless of spread — e.g. terms involving $A_i - \bar{A}_{\text{pool}}$ (discussion: $w(A_i - \bar{A}_{\text{pool}}) + (1 - w)A_i$ and algebraically related forms). Map to 537 score / weight construction before investing in **A**.

Separate exploratory idea (later): promotion may depend on **own ability × pool context** (high/mid/low A strata do not add up to the aggregate U-shape) — not the same as **A**, but related to interaction terms in the outcome model, not pool construction per se.

Where each thread lives (do not merge notebooks)

Layer	Pools	PD11 threads
538	Empirical: fixed panel (<code>team_id</code> , <code>season</code>). Generative (planned): soft assignment calibrated to 530	B (4B/4C) + §6 ladder; A when generative cells land
537	Frozen legacy sim — sort-and-chop + Cell 10 widgets	C (promotion score); compare old assignment only
530 CELL 5–9	Real panel forensics	Mean/SD histograms, mean×SD, interval overlap
535 / <code>tier_1_roadmap.md</code>	Pipeline + heavy EDA on same realized pools	Mechanism columns; not generative reassignment

Config files: `sports/sim_config.py` = 537 only. `sports/tier1_sim_config.py` = generative defaults for 538. Design note: `sports/documents/Tier1_Presorting_Design_Note.md`.

537 anchors (legacy, do not extend for Thread A): `assign_pool_ids`, `sorting_signal_for_pools`, `SORTING_NOISE_SD`; Cell 10 (`cell10_playground_run.py`, `cell10_knob_catalog.py`, `sim_config.py`).

How §6A relates to §7 and §9

- §6 / 538: minimal **reduced form** on **realized** pools (Wang replay) — still required for Alex conversations and cross-domain comparability.
- §6A / 538 (generative): new **assignment rules** calibrated via 530; 537 unchanged for comparison.
- §9: cross-domain program keeps the **same §6 ladder** per domain; §6A is the shared **assignment + promotion rulebook** to align Army / academia / sports simulations before claiming universality.

7. Where simulation fits (537 vs 538)

- 538 (empirical): §4–§6 ladder on **real** (`team_id`, `season`) pools — primary Alex deliverable.
- 538 (generative, planned): soft assignment + replay **530-style** overlap checks; knobs in `tier1_sim_config.py`.
- 537 (frozen): legacy sort-and-chop lab + Cell 10 widgets; use only to compare **old** assignment or to try **promotion-score** (Thread C) tweaks via `sim_config.py`.
- 530 CELL 5–9: forensic targets for calibrating generative assignment (not simulation).

8. Deliverables for the next conversation

- This outline filled in at **talking-point** depth (sub-bullets, not dissertation length).
- Matching cells in `538_alex_tier1_model_and_fit.ipynb` so figures/tables follow sections 4–6.

9. Wang-style program — same skeleton, many domains

Alex’s sequence is how you **lock a minimal empirical model** in one domain. The Wang-line aspiration in your notes is: **(i)** a stable empirical shape, **(ii)** a **small** generative story, **(iii)** extra **testable** predictions, **(iv)** replay in other domains.

Keep constant across domains (the “minimal model that does the trick”):

Layer	What stays the same	What changes per domain
Unit	Individual (or role-holder) embedded in a local pool at time t ; outcome is globally scarce.	Definition of pool (team, unit, department) and calendar.
Objects	Y , amalgamated L (peer/rivalry proxy), Λ_t , baseline A .	How you measure L , Y , Λ from logs.
Estimand	Turning point L^* (or equivalent in z-score then mapped back).	Binning / windowing rules; institutional noise.
Fitting ladder	Bins $\rightarrow$ quadratic in L (LPM) $\rightarrow$ logit/probit $\rightarrow$ one decomposition at a time.	FE structure (season vs fiscal year vs cohort).

How you *find* minimality (not wish it):

1. **Start with the briefing’s Tier 1 spec** — one L proxy, $L + L^2 + A$, report L^* . That *is* the candidate minimal reduced form.
2. **Add complexity only on forensic rules:** e.g. crowding or minutes enters only if (a) theory says the shape should move with that margin *and* (b) the data move that way, or the minimal spec is badly misspecified in a pre-registered sense. Your “one diagnostic at a time” rule is the operational version of minimality.
3. Use **537** for Wang-style *comparative statics* without bloating the empirical notebook: e.g. shift effective Λ , concentration, or opportunity; check whether L^* / top-bin behavior moves in the direction **Vector_Questions** lists (shift of peak, stronger top decline under congestion). That ties “simple mechanism” to **extra predictions**, not only the inverted-U picture.
4. **Per new domain:** copy the **same section headings** (this doc §1–§6) into a short domain addendum or notebook; replace §4’s table with that domain’s column mapping; rerun the ladder. The **abstract** story should read almost unchanged; only the measurement bridge changes.

Honest scope: “All domains” is **parallel replay of the same checklist**, not one pooled mega-regression on day one. The Wang papers earn universality by showing the **same qualitative mechanism** with domain-specific codings; your Alex outline is the contract for what must match across those codings.

10. **538** implementation map (keep in sync with notebook)

Notebook: `sports/538_alex_tier1_model_and_fit.ipynb`
Switchboard: CELL 0 (`RUN_CELL*`, `PERF_METRIC`, `PRIMARY_POOL_MODE`, `COMPUTE_WEIGHTED_CROWDING`, ...).

Outline §	Step / object	538 cells	Notes
§4	Data map	0 → 1 → 2 → 3	Same pipeline as 535 cells 1–3; no heavy 535 CELL 4 EDA.
§6.1	Descriptive bins	4, 4A	4 = table; 4A = binned draft rate vs mean L in bin.
§6A (PD11-B)	Q × peer SD diagnostic	4B, 4C	Associational; see §6A. Gated by RUN_CELL4B.
§6.2	Quadratic LPM + L^*	5	Overlay: binned means + fitted LPM ($\tilde{A} = 0$).
§6.3	Logit + L^*	6	Same z-scores as 5; needs statsmodels.
§6.4	Inference / robustness	7+	Not implemented yet.
§6A (PD11-A)	Generative pool assignment (planned)	538 + tier1_sim_config.py	Calibrate to 530 CELL 8–9.
§6A (PD11-C)	Promotion score experiments	537 Cell 10, sim_config.py	Legacy only; optional compare to 538.
§7	Forensics / legacy sim	530 CELL 5–9; 537 frozen	Empirical ladder in 538 §4–§6.

When you add or rename a 538 cell: update this table and the §6 bullets above.

What this document is *not*

- Not “PD11 = only the heatmap.” PD11’s center of gravity is pre-sorting / pool construction (§6A thread A, planned in 538; 537 frozen). The Q × peer SD block (4B / 4C, §6A thread B) is one empirical check; promotion vs rank (§6A thread C) can still be probed in 537 Cell 10.
- Not “538 solves pool formation.” 538 §6 assumes realized (team_id, season) pools; see §6A.
- Not the same as tier_1_roadmap.md. That file is the living checklist for 535 (pipeline contract, df columns, 535 CELL 0). 538 follows this outline §4–§6; the roadmap links here but still centers 535 for mechanism EDA and exports.

- Spine for Alex meetings: §1–§6 + §10 (538 empirical). Generative rulebook: §6A + `Tier1_Presorting_Design_Note.md` + `tier1_sim_config.py` (538, when coded). Legacy sim: 537 only.

Changelog

Date	Note
2026-05-18	<code>Tier1_Presorting_Design_Note.md</code> ; <code>tier1_sim_config.py</code> ; generative → 538, 537 frozen; <code>sim_config</code> 537-only.
2026-05-18	§6A: PD11 pool formation, promotion rules, 538 vs 537 split; §3/§7/§10/"What this is not" cross-refs.
2026-05-18	§10: 538 cell map; §4/§6 cross-refs; clarify vs <code>tier_1_roadmap</code> and Paper Directions 11.
2026-05-12	§9: Wang-style cross-domain program — same skeleton, measurement bridge per domain; 537 for comparative statics.
2026-05-12	Initial skeleton: Alex-sequenced spine; links to existing briefing + new 538 notebook.